

These are not, generally, mutually orthogonal, since $\mathbf{e}^\pm$ are not unit-length vectors. Also the magnetic fields of the modes 2, 3 lie in the common plane spanned by the two vectors $\mathbf{h}^\pm$ and are linearly combined of them with the same coefficients as in (38). It can be checked that the electric and magnetic fields in each mode are mutually orthogonal, of course: $(\mathbf{h}^{(2,3)}\mathbf{e}^{(2,3)}) = 0$.

In the special case of only one invariant different from zero, $\Lambda_3 \sim \mathfrak{G} \approx (\mathcal{E}/\mathcal{B}) \rightarrow 0$, the eigenvectors $\mathbf{b}_\mu^{(2,3)}$ (7) are the same as $c_\mu^\pm$ (11) owing to (10), hence $\mathbf{e}^\pm$, eqs. (35), (36), and $\mathbf{h}^\pm$, eq. (37), become the electric and magnetic fields of the corresponding eigenmodes, coinciding with their expressions known from Refs. [4, 14] with the particular property, that the electric field of mode 2 lies in the plane $(\mathbf{3}, \mathbf{2})$, while that of mode 3 is orthogonal to this plane, known for the special case of zero virtuality, $k^2 = 0$, from Ref. [16].

B. Magnetoelectric effect

In the magnetic-like field it is known that virtual photons of mode 2 are carriers of electrostatic [3, 25] forces, whereas those of modes 1 and 3 are responsible for magnetostatic interaction [3]. These statements follow from the representation (31) for the basic vectors, that become eigenvectors in that special case, and from the diagonal representation of the Green function (28) that allows to write electric and magnetic fields created by various (static included) configurations of small – as compared to the background field – charges and currents. The mixing of the basic vectors (10) in eigenmodes 2 and 3 for the general external field with $\mathfrak{G} \neq 0$ makes these statements no longer true in what concerns these modes, mode 3 remaining as it was. Moreover, thanks to the mixing, a static electric charge, if placed in the external field with the both invariants different from zero, gives rise not only to an electric field, as usual, but also to a magnetic field of its own, like a magnetic charge or moment. Also stationary currents produce some electric admixture to their customary magnetic fields.

Here we consider this analogue to the magneto-electric effect known in crystals [37] using the field of a point-like static charge q taken at rest in the special frame as an example. We set the 4-current corresponding to this source in the coordinate space x as $j_\mu(x) = q\delta_{\mu 0}\delta^3(\mathbf{x})$, where $\delta_{\mu 0}$ is the Kronecker symbol and $\delta^3(\mathbf{x})$ is the Dirac delta-function. Integrating this current with the Green function (28) we obtain for the vector-potential produced by the